

Applied Human Computer Interaction

Project Proposal

Designing User-Centered Brain–Computer Interfaces: Improving Feedback, Training, and Usability

Topic: Brain-Computer Interfaces (BCI)

Team Members

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1. Introduction

Brain–computer interfaces (BCIs) create a direct communication pathway between brain activity and an external computer or device. They can support communication, rehabilitation, assistive control, and new forms of interaction for people who cannot reliably use conventional input methods. Despite progress in signal decoding, many BCI systems remain difficult to use outside controlled settings. Users may need repeated calibration and training, receive feedback that is hard to interpret, or experience mental workload and frustration that are not captured by technical accuracy alone. This project investigates how a human-centered interface can make BCI interaction easier to understand, learn, and evaluate.

2. Problem Statement

BCI development is often reported through technical measures such as classification accuracy and information-transfer rate. These measures are important, but they do not fully explain whether users understand system state, know when a command has been recognized, recover from errors, or remain motivated during training. Existing work discusses user-centered design, feedback, workload, usability, and gamification, yet these factors are often examined in different application contexts and with different evaluation measures. The project will investigate whether a prototype that combines clear system-state feedback, guided training, user-adjustable settings, and error-recovery controls can improve perceived usability and task performance compared with a basic BCI-style interface.

3. Research Objectives and Questions

- Review recent BCI research on human factors, usability, feedback, training, accessibility, and user experience.
- Identify interface-level barriers that make BCI systems difficult to learn, interpret, or use consistently.
- Design a prototype that communicates system state clearly and supports guided practice, adjustable feedback, and recovery from misinterpreted commands.
- Evaluate the prototype using task performance, usability ratings, perceived workload, and participant feedback.

Research questions:

- RQ1: Which interaction barriers most affect users’ understanding and perceived usability of BCI systems?
- RQ2: How do feedback clarity and guided training affect users’ task completion, error recovery, and confidence?
- RQ3: Do user-adjustable feedback and recovery controls improve perceived usability compared with a basic BCI-style interface?
- RQ4: How can BCI interfaces be evaluated beyond signal accuracy, using measures of effectiveness, efficiency, workload, and satisfaction?

4. Selected Literature

The following sixteen publications from 2021–2025 provide an initial research base on human-centered BCI design, usability, feedback, training, accessibility, and ethics. The list includes journal and conference papers as well as a preprint; each item should be read in full and its bibliographic details confirmed before final submission.

# / Paper	Relevance to this project	Link
1. Lu et al. (2021), “Human factors engineering of brain-computer interface and its applications: Human-centered brain-computer interface design and evaluation methodology.”	Establishes a human-centered framework for BCI design and evaluation; supports measuring user needs and experience alongside technical performance.	https://doi.org/10.7507/1001-5515.202101093
2. Zulauf-Czaja et al. (2021), “On the way home: a BCI-FES hand therapy self-managed by sub-acute SCI participants and their caregivers: a usability study.”	Examines usability in a real rehabilitation context and highlights the importance of participant and caregiver needs.	https://doi.org/10.1186/s12984-021-00838-y
3. Haider (2021), “Grey Matter’s Grey Areas: Privacy in the Age of Brain-Computer Interfaces.”	Frames neural data as highly personal and motivates transparent, user-respecting control over BCI data.	https://philsci-archive.pitt.edu/24219/
4. The effect of visual and proprioceptive feedback on sensorimotor rhythms during BCI training (2022).	Compares feedback modalities during motor-imagery training; relevant to designing understandable	https://doi.org/10.1371/journal.pone.0264354

	feedback loops.	
5. “A systematic review of research on augmentative and alternative communication brain-computer interface systems for individuals with disabilities” (2022).	Reviews AAC-BCI evidence and highlights variability across users and the need for studies involving people with disabilities.	https://doi.org/10.3389/fnhum.2022.952380
6. Lyu et al. (2023), “Human factors engineering of BCI: an evaluation for satisfaction of BCI based on motor imagery.”	Uses satisfaction, workload, sensor ratings, and interviews, supporting a multi-measure evaluation plan.	https://doi.org/10.1007/s11571-022-09808-z
7. “Paving the Way for Motor Imagery-Based Tele-Rehabilitation through a Fully Wearable BCI System” (2023).	Explores wearable EEG and extended-reality feedback across repeated sessions, relevant to comfort and practical use.	https://pmc.ncbi.nlm.nih.gov/articles/PMC10346666/
8. Arpaia et al. (2023), “Visual and haptic feedback in detecting motor imagery within a wearable brain-computer interface.”	Investigates visual and haptic feedback in a wearable BCI context, informing multimodal feedback design.	https://doi.org/10.1016/j.measurement.2022.112304
9. Atilla, Alimardani, and Postma (2023), “Improving User Experience and Performance through Gamification of MI-BCI Training.”	Discusses game elements for more engaging motor-imagery training and the role of motivation and attention.	https://doi.org/10.3217/978-3-85125-962-9-40
10. “Privacy and Ethical Concerns of Brain-Computer Interfaces” (IEEE MetaCom, 2023).	Examines privacy and ethics in current and prospective BCI applications; informs consent and data-transparency requirements.	https://doi.org/10.1109/MetaCom57706.2023.00036
11. Sun and Ye (2023), “The functional differentiation of brain-computer interfaces (BCIs) and its ethical implications.”	Argues that ethical governance should account for differences between BCI functions, supporting context-sensitive design.	https://doi.org/10.1057/s41599-023-02419-x
12. “Comprehensive evaluation methods for translating BCI into practical applications: usability, user satisfaction and usage of online BCI systems” (2024).	Reviews evaluation beyond offline algorithm performance, including usability, satisfaction, and system-user fit.	https://doi.org/10.3389/fnhum.2024.1429130
13. “Evaluating the real-world usability of BCI control systems with augmented reality: a user study protocol” (2024).	Proposes a staged user-centered protocol combining technical validation, task performance, and user-experience measures.	https://doi.org/10.3389/fnhum.2024.1448584
14. Atilla, Postma, and Alimardani (2024), “Gamification of motor imagery brain-computer interface training protocols: A systematic review.”	Synthesizes 86 studies and reports that feedback, avatars, assistance, and social interaction can support training experience.	https://doi.org/10.1016/j.chbr.2024.100508
15. Qian and Tanaka (2024), “Personalized SSVEP-BCI Text Spelling User Interface.”	Proposes a keyboard-like stimulation layout with grouped letters, relevant to personalization and reducing interface complexity.	https://doi.org/10.5057/isase.2024-c000003
16. “Evolution of users’ subjective experience over three training sessions with an EEG Motor-Imagery Brain-Computer Interface (MI-BCI)” (2025).	Tracks subjective experience across sessions and reports that workload and fatigue can change over time, motivating repeated-session evaluation.	https://www.sciencedirect.com/science/article/pii/S0006899325006481

5. Preliminary Research Gap

A detailed gap analysis will follow full-text reading and comparison of the selected papers. Based on the initial review, the project will investigate the following possible gaps:

- Many BCI studies report decoding accuracy and information-transfer rate, while the user-facing experience—understanding system state, confidence in recognized commands, workload, and error recovery—may receive less consistent measurement.
- Research has investigated feedback modalities, gamified training, wearable systems, and usability evaluation, but these approaches are often assessed in different settings and are not always combined in a single interface comparison.
- A focused HCI comparison of a basic BCI-style interface against one combining clear state feedback, guided practice, and user-controlled recovery may help clarify which interface features improve usability and perceived workload.

6. Expected Deliverables

- Structured literature review of at least fifteen papers published between 2021 and 2026.
- Problem statement, research questions, and preliminary research-gap analysis.
- Figma prototype showing BCI-style command selection, system status, guided practice, feedback, and error recovery; the prototype will simulate decoded commands and will not claim to read brain signals.
- Small user study comparing the proposed design with a baseline interface, subject to course and participant constraints.
- Analysis of task completion, errors, perceived workload, usability ratings, and qualitative feedback.
- Final research report formatted in IEEE style.

7. Tentative Timeline

Phase	Activities
Weeks 1–2	Read and annotate the selected papers; refine the research questions and gap.
Weeks 3–4	Define user tasks, evaluation measures, and prototype requirements; sketch wireframes.
Weeks 5–6	Build the Figma prototype and baseline; pilot test task wording and questionnaire.
Weeks 7–8	Conduct the user study, collect data, and analyze usability and workload measures.

Weeks 9–10	Write the IEEE report, revise the literature review and limitations, and submit.
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References

- [1] X. Lu et al., “Human factors engineering of brain-computer interface and its applications: Human-centered brain-computer interface design and evaluation methodology,” *Sheng Wu Yi Xue Gong Cheng Xue Za Zhi*, vol. 38, no. 2, pp. 210–223, 2021, doi: 10.7507/1001-5515.202101093.
- [2] A. Zulauf-Czaja et al., “On the way home: a BCI-FES hand therapy self-managed by sub-acute SCI participants and their caregivers: a usability study,” *J. NeuroEngineering Rehabil.*, vol. 18, art. 44, 2021, doi: 10.1186/s12984-021-00838-y.
- [3] S. Haider, “Grey Matter’s Grey Areas: Privacy in the Age of Brain-Computer Interfaces,” preprint, 2021. [Online]. Available: <https://philsci-archive.pitt.edu/24219/>.
- [4] “The effect of visual and proprioceptive feedback on sensorimotor rhythms during BCI training,” *PLOS ONE*, 2022, doi: 10.1371/journal.pone.0264354.
- [5] “A systematic review of research on augmentative and alternative communication brain-computer interface systems for individuals with disabilities,” *Front. Hum. Neurosci.*, 2022, doi: 10.3389/fnhum.2022.952380.
- [6] X. Lyu et al., “Human factors engineering of BCI: an evaluation for satisfaction of BCI based on motor imagery,” *Cogn. Neurodyn.*, vol. 17, pp. 105–118, 2023, doi: 10.1007/s11571-022-09808-z.
- [7] “Paving the Way for Motor Imagery-Based Tele-Rehabilitation through a Fully Wearable BCI System,” 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10346666/>.
- [8] P. Arpaia et al., “Visual and haptic feedback in detecting motor imagery within a wearable brain-computer interface,” *Measurement*, vol. 206, art. 112304, 2023, doi: 10.1016/j.measurement.2022.112304.
- [9] F. Atilla, M. Alimardani, and M. Postma, “Improving User Experience and Performance through Gamification of MI-BCI Training,” in *Proc. 10th Int. Brain-Computer Interface Meeting*, 2023, doi: 10.3217/978-3-85125-962-9-40.
- [10] “Privacy and Ethical Concerns of Brain-Computer Interfaces,” in *Proc. IEEE Int. Conf. Metaverse Computing, Networking and Applications (MetaCom)*, 2023, doi: 10.1109/MetaCom57706.2023.00036.
- [11] X.-Y. Sun and B. Ye, “The functional differentiation of brain-computer interfaces (BCIs) and its ethical implications,” *Humanities and Social Sciences Communications*, vol. 10, art. 878, 2023, doi: 10.1057/s41599-023-02419-x.
- [12] H. Pan et al., “Comprehensive evaluation methods for translating BCI into practical applications: usability, user satisfaction and usage of online BCI systems,” *Front. Hum. Neurosci.*, vol. 18, art. 1429130, 2024, doi: 10.3389/fnhum.2024.1429130.
- [13] “Evaluating the real-world usability of BCI control systems with augmented reality: a user study protocol,” *Front. Hum. Neurosci.*, 2024, doi: 10.3389/fnhum.2024.1448584.
- [14] F. Atilla, M. Postma, and M. Alimardani, “Gamification of motor imagery brain-computer interface training protocols: A systematic review,” *Comput. Human Behav. Rep.*, vol. 16, art. 100508, 2024, doi: 10.1016/j.chbr.2024.100508.
- [15] Q. Qian and H. Tanaka, “Personalized SSVEP-BCI Text Spelling User Interface,” 2024, doi: 10.5057/isase.2024-c000003.
- [16] “Evolution of users’ subjective experience over three training sessions with an EEG Motor-Imagery Brain-Computer Interface (MI-BCI),” *Brain Research*, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0006899325006481>.

Note: The prototype is planned as an HCI interface simulation, not a medical device or a functioning neural-signal decoder. Any human study will use informed consent and non-clinical tasks, and will report limitations transparently.